Task 3 – Ledger Duplicate-Customer Fix Specification

Developer-ready specification for implementation with Claude Code

1. Problem

The Ledger test migration produced 1,200 duplicate customer records while migrating data from the legacy CRM. Different representations of phone numbers (for example, +91, a national leading zero, spaces or punctuation) may cause the same number to be treated as different values; this is a hypothesis to validate, not an established root cause. Implement deterministic normalization and conservative duplicate detection without merging distinct people.

2. Matching and data-handling rules

- Retain an immutable copy of each source row and source-system identifier; never overwrite the source data.
- Normalize phone values for comparison only: trim whitespace; remove display separators such as spaces, hyphens and parentheses; parse country/national prefixes using an approved India numbering rule/library. Do not blindly strip arbitrary digits or assume every number is valid.
- Treat +91XXXXXXXXXX and 0XXXXXXXXXX as equivalent only when parsing confirms both represent the same valid Indian national number. Store normalized E.164 form separately from the original value.
- Use a hierarchy: exact stable legacy customer ID is strongest; otherwise use a validated normalized phone plus a second corroborating attribute (for example verified email or approved name/date-of-birth combination). The client must approve the hierarchy.
- Phone match alone is not sufficient for an automatic merge where names or other identity attributes conflict, where a phone is shared, or where the number is invalid/ambiguous. Send these cases to manual review.
- Do not fuzzy-merge on name alone. Define approved case/diacritic/whitespace normalization for names only as a supporting signal.
- Choose a canonical record using a client-approved survivorship rule (for example, trusted source priority and most recently verified non-empty fields). Preserve all source IDs and field-level provenance.
- Make the job idempotent: rerunning the same source data must not create further duplicates. Produce an audit log of candidate pairs, rule matched, decision, canonical ID and rejected/merged IDs.
- Run in dry-run mode first; export proposed merges and counts for business approval. Back up the target and provide a tested rollback before applying merges.

3. Acceptance criteria – Given / When / Then

Given	When	Then
Two records have the same valid phone represented as +91 9876543210 and 09876543210, and a second approved identity attribute agrees	The normalization and matching job runs	They are grouped as a duplicate candidate; merge only under the approved policy and retain provenance.
Two records have different validated phone numbers and different legacy IDs	The job runs	They remain separate customers.
Two records share a phone but have conflicting names or evidence of a shared household phone	The job runs	No automatic merge occurs; both are flagged for manual review.
A phone value is malformed or cannot be parsed confidently	The job runs	Original value is preserved; record is flagged, not guessed or auto-merged.
A dry-run has produced candidate groups	An authorized reviewer approves a group	The merge applies the approved survivorship rule and records an audit trail.
The same source file is processed twice	The migration is rerun	The second run creates no additional duplicate customers and produces consistent counts.
A merge has been applied	Rollback is invoked using the recorded backup/audit data	The target can be restored to the pre-merge state and reconciliation is possible.

4. Small test-data table

ID	Phone	Other evidence	Expected result
A	+91 98765 43210	Verified email same as B	Candidate match with B after valid normalization; merge only if policy permits.
B	09876543210	Verified email same as A	Candidate match with A; confirm national prefix parsing.
C	+91 91234 56780	Different legacy ID and verified email	Separate customer.
D	+91 98765 43210	Name conflicts with A; shared phone possible	Manual review; no automatic merge.
E	98-76-ABC	No corroborating identifier	Invalid/ambiguous; preserve and flag.

5. Client questions before work starts

- Which field is the authoritative unique customer identifier, and is phone number unique or can it be shared?
- Which phone formats/countries are present? Can the client provide valid examples and expected normalized values?
- Which secondary fields may be used for matching (verified email, name, date of birth, address), and which are prohibited?
- Which source wins when fields conflict? What are the canonical-record and field-level survivorship rules?
- Must inactive/deceased/closed accounts remain distinct or be retained under special rules?
- Who approves manual-review groups and production merges? What audit retention and privacy controls apply?
- What are the reconciliation tolerances, backup/rollback expectations and migration acceptance thresholds?

Implementation note: do not use phone-number equality as a sole merge key. Validate numbering rules and identity policy with Sagar Publications before applying changes.